





Training for Wilmar Group
 "Selection Guide of Radar Level
 Measurement"

Gresik, 12th December 2013.

Endress+Hauser 
 People for Process Automation

Slide 1 / 37

Training

Products & Services made by Endress+Hauser

Level	Flow	Analysis	Pressure	Temperature	Recorder	System Component	Tank Gauging
Capacitance	Electro magnetic	pH	Absolute	RTD	Paper	Display	Servo
Conductive	Vortex	Conductivity	Gauge	Thermocouple	Hybrid	Power supply	Radar
Hydrostatic	Coriolis	Turbidity	Differential	Transmitter	Paperless	Limit Contacter	Float
Differential	Ultrasonic	Chlorine	Hydrostatic		Data logger	Surge Arrester	Tank Computer
Vibration	Thermal Mass	Dissolved Oxygen				Communication	
Ultrasonic	Delta press.	Sludge Level					
Microwave	- Venturi	Ammonium					
Micro impulse	- Orifice	Silicate					
Electromech	- Pitot	COD					
Radiometric	- Nozzle	BOD					
Servo							

The Level product range

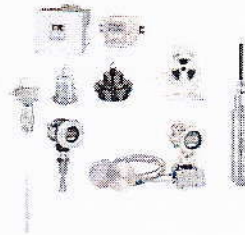
Pressure



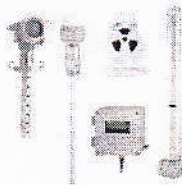
Level limit detection



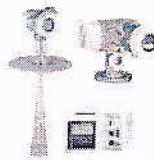
Continuous level



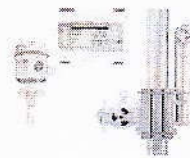
Interface



Tank gauging



Density



Slide 3

Endress+Hauser

Tof Products

Time of Flight - the new generation

FMR240



- Horn: DN40 ... DN100
- Material: 316L Alloy C22
- PTFE cone + O-ring
- G 1 1/2", 1 1/2" NPT, DN50...DN150, ANSI, JIS
- Vacuum ... 64 bar
- Visc: -40°C ... 150°C *)
- Kältes: -20°C ... 150°C
- *) 200°C for Wave Guide

FMR245



- Horn: DN40, DN80
- Material: PTFE (TFM1500)
- flush plated 4mm
- DN50...DN150, ANSI, JIS

- Vacuum ... 16 bar
- -40°C ... 150°C

FMR244



- Horn: DN40
- Material: PVDF + PTFE
- Viton O-ring
- G 1 1/2", 1 1/2" NPT

- Vacuum ... 3 bar
- -40°C ... 130°C

FMR 5x Series



Folie 4

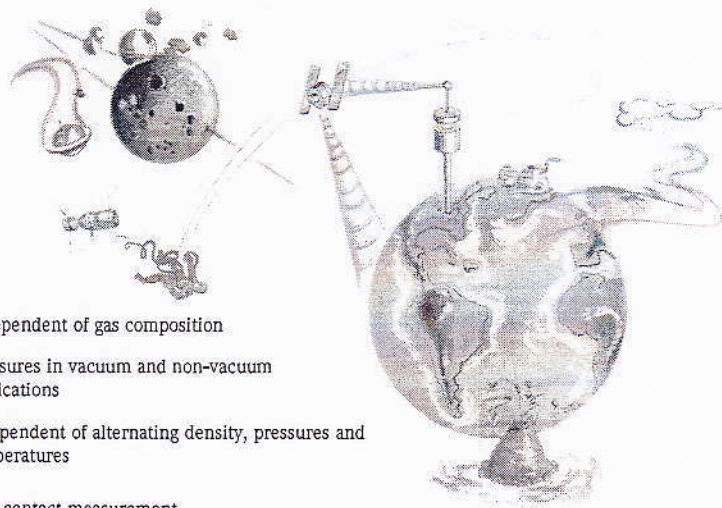
Endress+Hauser

Tof Products

Micropilot M

Benefits of electromagnetic waves

- Independent of gas composition
- Measures in vacuum and non-vacuum applications
- Independent of alternating density, pressures and temperatures
- Non-contact measurement
- Independent of air turbulences

Endress+Hauser 

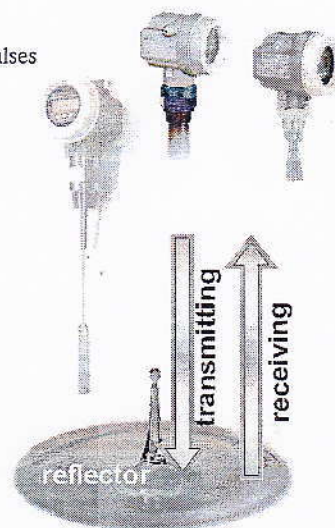
Tof Products

Time of Flight

Measuring Principle

- Transmission of ultrasonic or microwave impulses
- Reflection on the surface of the medium
- Receiving of the reflected impulses
- Calculation of the distance between unit and measured surface via:

$$d = c \cdot \frac{t}{2}$$

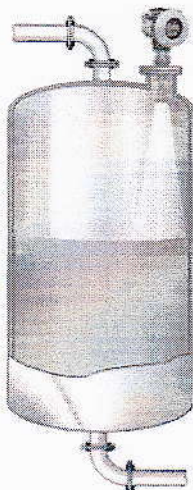


Folie 6

Endress+Hauser 

Tof Products

Radar level measurement



Radar

Micropilot works with high-frequency radar pulses which are emitted by an antenna and reflected from the product surface. The time of flight of the reflected radar pulse is directly proportionate to the distance traveled. If the tank geometry is known, the level can be calculated from this variable.

Micropilot M

Non-invasive, maintenance-free measurement also under extreme conditions. Unaffected by density, temperature, conductivity and humidity. No impairment by outgassing media (steam pressure).

Process temperatures up to +400 °C

Process pressures up to 160 bar

Folie 7

Endress+Hauser

Tof Products

Guided Wave Radar level measurement



Guided radar

Levelflex M works with high-frequency radar pulses which are guided along a probe. As the pulses impact the medium surface, the characteristic impedance changes and part of the emitted pulse is reflected. The time between pulse launching and receiving is measured and analyzed by the instrument and constitutes a direct measure for the distance between the process connection and the product surface.

Levelflex M

Reliable and maintenance-free measurement in liquids, also in turbulent media and foam. Unaffected by density, temperature, conductivity and humidity. No impairment by outgassing media (steam pressure).

Process temperatures up to +400 °C

Process pressures up to 400 bar

Folie 8

Endress+Hauser

Tof Products

Time of Flight

Mechanical / Electromagnetical waves

	Frequency	Wave length λ	
Telephone	1 Hz	kilometre	
AC current	10 Hz		
	100 Hz		
Radio	1 kHz		
Ultrasound	10 kHz		
	100 kHz	metre	
Television	1 MHz		
	10 MHz		
	100 MHz - (10 Hz)		
Microwaves	1 GHz	decimetre	
	10 GHz	centimetre	
	100 GHz	millimetre	
Heat	1 THz - (10 ¹² Hz)	micrometer	
	10 THz		
	100 THz		
visible Light	1 - (10 ¹⁵ Hz)	nanometre	
	10		
ultra violet	1 - (10 ¹⁶ Hz)		
	10		
	100		
X-rays	1 - (10 ¹⁹ Hz)	picometer	
	10		
	100		
Gammawaves cosmic radiation	1 - (10 ²⁶ Hz)	femtometer	
	10		
	100		

Prosonic:
approx. 40 ... 70 kHz
(at 0°C)
(mechanical waves)

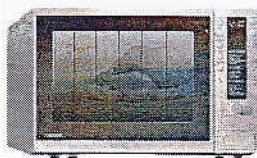
Levelflex
approx. 200 MHz ... 1,5 GHz
(electromagnetic waves)

Micropilot:
FMR 53x: approx. 6 GHz
FMR 23x: approx. 6 GHz
FMR 24x: approx. 26 GHz
(electromagnetic waves)

Endress+Hauser

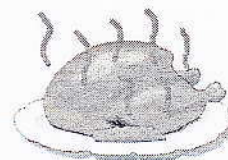
Tof Products

Radars waves - Effects on humans ?



Inside: app. 1 W/cm²

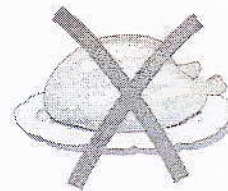
Leakage at closed door:
app. 5 mW/cm²



At transmitting power of 2 W:
app. 0,1 mW/cm²



Pulse during measurement:
app. 0,00021 mW/cm²



Tof Products

Time of Flight

The M-Platform

IP 68 / NEMA 6P

Local display +
Menu driven set upPulseMaster® 2.0
Analysis (envelope) on site

Approvals



ATEX

PROFIBUS

HART

Fieldbus

Remote operating
and display

ToF-Tool + Documentation

ress+Hauser

Folie 11

Mettler

Tof Products

Set-up

Display

- Large 4-line display
- Easy user guidance
- Intuitive operation
- Help function

Easy and fast -
yet correct set-
up

ToF-Tool

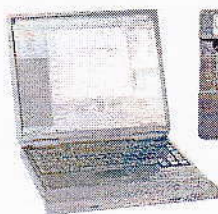
- Same operating concept as local display
- Graphical support of the commissioning steps
- Online help function (manual, ...)
- Documentation incl. envelope curve

Folie 13

Endress+Hauser

Tof Products

Analysis and Optimization



Reliable
measurement

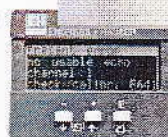
Predictive Reliability

- Ideal echo curve
- Indication of echo quality

Analysis and Optimization

- Envelope curve – lets you see what Levelflex M sees

Trouble Shooting



- Clear error description
- Hints for trouble shooting

Folie 14

Endress+Hauser

Competence in Power & Energy

Selection Step by step

This selection provides information on different measuring principles for continuous level/interface measurement in liquids as well as application and installation.

1. Overview of measuring principles
2. Checklist of the application requirements for the correct selection of a suitable instrument, it provides an overview and is supposed to help you to
3. Selection of the measuring principle, the appropriate measuring principle is first selected according to the application and its criteria (tank, bypass, stilling well, etc.). Select the principle which meets, if possible, all of the criteria required by you or your plant. The measuring principles are classified according to „non contact“ and „contact“ criteria.
4. Instrument selection - Compare your application and process data with the instrument data.

Slide 15

Endress+Hauser

Overview of measuring principles

						
Micropilot Non-contact, maintenance-free measurement also under extreme conditions. Unaffected by: density, temperature, conductivity and humidity. No impairment by vapor pressure.	Leveflex Reliable and maintenance-free measurement in liquids, also in turbulent media and foam. Unaffected by density, temperature, conductivity and humidity. No impairment by vapor pressure. Measurement of interface and level.	Prosonic 37 mA/T Non-contact and maintenance-free measurement without impairment by product properties, e.g. dielectric constant, conductivity, density or humidity.	Cerabar, Deltapilot Unaffected by dielectric constant, foam, turbulence and obstacles. Condensate-proof, watertight and long-term stable.	Deltabar Unaffected by dielectric constant, foam, turbulence and obstacles. High overload resistance.	Liquicap Exact measurement from the end of the probe to the process connection without any blocking distance. Very fast response times. Unaffected by density, turbulence and vapor pressure.	Gammapiilot M Non-contact measurement from outside for all extreme applications, e.g. very corrosive.
• Process temperatures up to +450°C/+842°F • Process pressures up to 160bar/2320psi	• Process temperatures up to +450°C/+842°F • Process pressures up to 400bar/5,800psi	• Process temperatures up to +150°C/+302°F • Process pressures up to 3bar/44psi	• Process temperatures up to +400°C/+752°F	• Process temperatures up to +400°C/+752°F • Process pressures up to 420bar/6,090psi • Unaffected by ambient	• Process temperatures up to +200°C/+392°F • Process pressures up to 100bar/1,450psi	• Unaffected by media • Any process temperature • Any process pressure • Unaffected by gammagraphy

Slide 16

Endress + Hauser 

Overview of measuring principles

						
Process temperature -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process pressure -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process temperature -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process pressure -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process temperature -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process pressure -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi	Process temperature -70...+450°C -101...+842°F -1...+160bar -14.5...+2,090psi
Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi	Measuring range 0.1...100mV, 0.1...100psi
Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%	Instrument accuracy ±0.1% ±0.1% ±0.1% ±0.1%
Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation	Function may be affected by • Foam • Extreme turbulence, boiling • Corrosive build-up on • Strong interface formation
Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)	Accuracy may be affected by • Wall effects • Insulation reflections • Signal-to-noise ratio • Extreme pressure changes • e.g. 1.25 in a 20bar/29psi (1.25 in a 20bar/29psi)
Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below	Application limits Measurement up to 100mV, 0.1...100psi DC < 1.4 Lateral installation or from below

• 0.1% of span, control circuit

• 0.1% of span, 0.1% of span

• Measurement only up to the probe tip

• Measurement is possible up to the breaking distance (BD) of the sensor

• TK = Temperature coefficient

Slide 17

Endress + Hauser 

E+H Radar Tank Gauging Custody Transfer & Inventory



Fields of business Inventory Control vs. Custody Transfer



Inventory Control	Custody Transfer
Measurement of the content in a plant (How rich are we?)	For trading products or tax payment evaluation
API recommendations for level instrumentation:	API recommendations for level instrumentation:
Under factory conditions: +/- 3mm	Under factory conditions: +/- 1mm
In application: +/- 25mm	In application: +/- 4mm
No approval	OIML rules + Local approvals e.g. PTB/NMI



2.1.1 API MPMS Chapter 3.1B

3.1B.4.3.2 Calibration Prior to Installation (Factory Calibration)

The reading of an ATG to be used for custody transfer application should agree with a certified measurement instrument within ± 1 mm or ($\pm 1/16$ inch) over the entire range of the ATG. The certified measurement instrument should be traceable to the national standards and should be provided with a calibration correction table. The uncertainty of the reference should not exceed 0.5 mm or ($1/32$ inch), with the calibration correction applied.

The reading of an ATG to be used for inventory application should agree with a certified measurement instrument within ± 3 mm ($\pm 1/8$ inch) over the entire range of the ATG. The certified measurement instrument should be traceable to the national standards and should be provided with a calibration correction table.

*That means roof (GRH!),
and vapor influence*

3.1B.4.3.3 Error Caused by Installation and Operating Conditions

The total error of an ATG in custody transfer service should not be affected by more than ± 3 mm ($\pm 1/8$ inch) due to installation, to variation of operating conditions (refer to Section 3.1B.4.3.5) or variation of physical and electrical properties of the liquid and/or vapor, provided that these conditions are within the limits specified.

3.1B.4.3.4 Overall Accuracy of the Installed ATG

The overall accuracy of the installed ATG includes both the intrinsic accuracy of the ATG, as verified by factory calibration, and those effects caused by installation and operating conditions. The overall accuracy of an ATG in custody transfer service should be within ± 4 mm ($\pm 1/16$ inch). The overall accuracy of an ATG in inventory control service should be within ± 25 mm (± 1 inch).

*But what is the accuracy
of the tape?*

Slide 34 Ngo

Tank Gauging Solutions with Endress+Hauser

Manual dipping in crude oil

3.1A.9.1.1 Manual gauging shall require obtaining either two consecutive gauge readings to be identical, or three consecutive readings within a range of 3 mm ($1/8$ in.). If the first two readings are identical, this reading shall be reported to the nearest 1 mm if metric tapes are used, or to the nearest $1/8$ in. if customary tapes are used. When three readings are taken, all three readings shall be within the 3 mm ($1/8$ in.) range and readings averaged to the nearest 1 mm for metric tapes and $1/8$ in. for customary tapes.

For crude oil lease tanks of 1,000 bbl nominal capacity or less, the range may be increased to 5 mm (or $1/4$ in.), and should be reported to the nearest 5 mm (or $1/4$ in.).

Slide 35 Ngo

Endress+Hauser 

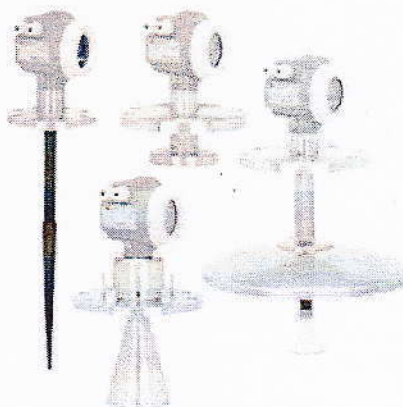
Endress+Hauser Radar Offering

- Two series (three levels of accuracy):
 - Micropilot S-series (high accuracy)
 - Micropilot S-series (inventory control versions)
 - Micropilot M-Series
- High accuracy Micropilot S-series is approved for Custody Transfer applications
 - Instrument accuracy : better than ± 1 mm
 - Weights & Measures approved by NMI and PTB and other national authorities
 - OIML certified
- Inventory control Micropilot S-series
 - Instrument accuracy: ± 3 mm
 - Choice of antennas
 - Cost-effective solution
- Micropilot M-Series is well suited for inventory control applications:
 - Two-wire system
 - Instrument accuracy ± 3 to 10 mm (depending on type)
 - cost effective solution
- All have standard HART protocol and have the same User Interface
- Integration with Tank Side Monitor NRF 590

Endress+Hauser 

Micropilot S-Series

- For HIGH ACCURACY and INVENTORY CONTROL
- Four different versions:
 - FMR 540 – Horn antenna
 - FMR 531 – Rod antenna
 - FMR 532 – Planar antenna
 - FMR 533 – Parabola antenna



Endress+Hauser 

Radar Selection Criteria

- What is your application? (I.e. tank type?)

- What accuracy is required?

- Inventory vs. custody transfer?

- What type of product is in the tank

- **Dielectric constant**

The table describes the classification of products with different dielectric constants. In cases where the dielectric constant of a product is not known, we recommend that product class B is assumed to ensure reliable measurement.

Product class (PC)	DC (ε _r)	Examples
A	1.4 ... 1.9	Non-conductive liquids, e.g. LPG ¹
B	1.9 ... 4	Non-conductive liquids, e.g. white products, black products, crudes, bitumen/asphalts
C	4 ... 10	Chemicals, e.g. concentrated acids, organic solvents, esters, aniline, alcohol, acetone, etc.
D	>10	Chemicals (conductive liquids), e.g. aqueous solutions, dilute acids and alkalis

¹Treat ammonia (NH₃) as a product of class A, i.e. always use a stilling well

- What are the installation restrictions?

- Nozzle size - what nozzles are available and are they suitable?

- Tank type - Is your tank a cone roof, floating roof, bullet, etc.?

- What is the height of the tank? (I.e. what measuring range is required?)

- What are the application limits?

- What additional technical details must be considered? (I.e.

- temperature, pressure, etc.?)

Endress+Hauser 

Tank Gauging Applications

- Fixed roof tanks

- Including Asphalt applications within this type of tank

- Internal floating roof tanks

- With and without stilling well

- External floating roof tanks

- With and without stilling well

- Horizontal cylindrical tanks (non pressurized)

- FMR 240 radar with wave guide recommended

- Bullet tanks (pressurized)

- Including LPG applications within this type of tank

- Spheres

- Including LPG applications within this type of tank

- Severe duty applications

- High temperature, aggressive products, etc.

Endress+Hauser 

Fixed Roof High accuracy

Applications

FMR 530

- For restricted mounting positions
- Effective for tanks with small openings or flanges



FMR 531

- For aggressive products
- Effective for tanks with small openings or flanges
- (Stainless) heated feed lat applications



High Accuracy

Without stilling well

FMR 533

- Ideal for high precision measurements
- Ideal for petrol applications



With stilling well

FMR 532

- Ideal for high precision measurements in stilling wells
- Specifically designed for stilling well applications



Accuracy

Max. measure range (antenna size)	PC	DN 150 6"	DN 200/250 8"/10"	Roof	Parabolic antenna DN150 19"	Planar antenna DN150...DN300 (6"...12")
A						
B		15m (49ft)	20m (65ft)	10m (33ft)	1...40m (131ft)	1...38m (125ft)
C		15m (49ft)	20m (65ft)	15m (49ft)	1...40m (131ft)	1...38m (125ft)
D		20m (65ft)	25m (82ft)	20m (65ft)	1...40m (131ft)	1...38m (125ft)

Application limits

- Max. measuring range with custody transfer approval $\approx 25m$
- Strong formation of foam $\rightarrow$ Servo
- Hydrostatic tank gauging
- Check nozzle size - nozzle diameter $\geq$ DN150 / 6"
- FMR 531
- Use the antenna extension

- Max. measuring range with custody transfer approval $\approx 10m$
- Strong formation of foam $\rightarrow$ Servo
- Hydrostatic tank gauging

- Max. measuring range with custody transfer approval $\approx 25m$
- Blocking distance $\approx 1m$ (3.2ft)
- Strong formation of foam $\rightarrow$ Servo
- Hydrostatic tank gauging
- Mounting in manway not possible
- FMR 533

- Max. measuring range with custody transfer approval $\approx 20m$
- Pipe diameter $\geq 150mm$ / 6"
- FMR 530 with 3" or 4" hose

Endress+Hauser

Fixed Roof Inventory control

Inventory Control

FMR 530, 531, 532 and 533 Microplot S inventory control versions are available as optional accessories to satisfy your inventory control needs.

FMR 240

- Ideal for level measurements especially in small tanks
- Effective for secondary overflow protection



FMR 230

- For high pressures and temperatures
- Effective for tanks with small openings or flanges
- Effective for secondary overflow protection



FMR 231

- Cost effective solution for aggressive products
- Effective for tanks with small openings or flanges
- (Stainless) heated feed lat



Accuracy

Max. measure range (antenna size)	PC	DN 40 1.5/2"	DN 50 2"	DN 80 3"	DN 100 4"	DN 150 6"	DN 200/250 8"/10"	Roof
A								
B								
C								
D								

Application limits

- Please refer to the antenna size and maximum measuring ranges for the high accuracy versions of the Microplot S readers.
- Points / bubbles of surface $\rightarrow$ FMR 230 / FMR 231
- Atresina $\rightarrow$ FMR 230 with stilling well
- Heavy build up $\rightarrow$ FMR 230 with air purge

- Check nozzle size - nozzle diameter $\geq$ DN150 / 6"
- FMR 240 - FMR 231
- Use the antenna extension

- Servo $\rightarrow$ FMR 230 with stilling well
- Nozzle height $\approx 250mm$ $\rightarrow$ FMR 240 / FMR 231
- Heavy build up $\rightarrow$ FMR 230 with air purge

Endress+Hauser